

SMART CAMPUS SAFETY SYSTEM

CHRIST UNIVERSITY (KENGERRI CAMPUS) — COMPLETE TECHNICAL & FUNCTIONAL DOCUMENTATION

Version: 9.0 (v9) | Integrated Dataset: 57 Official Students | Primary Hotline: +91 8639527123

1. Executive Summary & Architecture Overview

The **Smart Campus Safety System** is an enterprise-grade, real-time safety, emergency dispatch, and incident management portal specifically engineered for **Christ University Kengeri Campus**. It provides instantaneous multi-channel emergency protection connecting students, campus security officers (Security Guard NNP), and university administration.

Core Architecture & Subsystems:

- Frontend Engine:** React 18 single-page application built with Vite bundler.
- User Interface Styling:** Tailwind CSS with custom dark glassmorphism theme (#070814 deep obsidian background).
- Vector Icons:** Lucide React Icons.
- Browser WebRTC Hardware Integration:** HTML5 MediaDevices API (navigator.mediaDevices.getUserMedia) for crisp camera photo capture and video proof recordings.
- Geolocation API:** Pinpoints exact live GPS coordinates (lat, lng) during emergency activations.
- Persistence Layer:** Browser LocalStorage versioning (scs_incidents_v9, scs_students_v9, scs_sos_logs_v9, scs_active_session_v9).
- State Synchronization:** Rapid 1.5-second polling loop ensuring real-time step advancement sync between Security Officer Control Desk and Student Dashboards.

2. Complete Official Student Dataset (57 Roster Records)

The system pre-seeds and maintains the complete roster of **57 official students** belonging to **4BTAD - AI & DS** (B.Tech 2024-2028). Each student record contains all 57 database fields, ensuring 100% data consistency across authentication, incident logging, SOS dispatch, and penalty management.

Default Password Computation Algorithm (computeDefaultPassword):

To eliminate login failures for students with single-letter initials (e.g. 'R JERPHIN', 'M ELIJO PAUL'), the system uses a smart password generator:

- Splits full student name into whitespace-separated name parts.
- Ignores single-letter initials (parts with length < 3).
- Extracts the first 3 letters of the primary name word (e.g., **JINKALA SHAHANAZ** → **JIN**, **R JERPHIN** → **JER**, **M ELIJO PAUL** → **ELI**).
- Prompts student to set a custom password on first login (isFirstLogin: true).

Sl.No	Reg No	Student Name	Class	Official Mail-Id	Registered Phone	Default Pass
1	2462002	A DARSHAN REDDY	4BTAD - AI & DS	a.reddy@btech.christuniversity.in	7338304910	DAR
2	2462003	A. B. CHARAN VYAS	4BTAD - AI & DS	a.vyas@btech.christuniversity.in	9980838384	CHA
3	2462005	ABHINAV ROY	4BTAD - AI & DS	abhinav.roy@btech.christuniversity.in	9717765104	ABH
4	2462006	ABRAHAM MODI S	4BTAD - AI & DS	abraham.s@btech.christuniversity.in	8296766468	ABR
5	2462008	ADITYA RAJ SHARMA	4BTAD - AI & DS	aditya.sharma@btech.christuniversity.in	7488092284	ADI

6	2462009	ADRITO GHOSH	4BTAD - AI & DS	adrito.ghosh@btech.christuniversity.in	9874052324	ADR
7	2462010	AHMAD HUSSAIN SAJID	4BTAD - AI & DS	ahmad.sajid@btech.christuniversity.in	9113693282	AHM
8	2462011	AKASH PATRA	4BTAD - AI & DS	akash.patra@btech.christuniversity.in	9382601713	AKA
9	2462012	ALBY SANTHOSH	4BTAD - AI & DS	alby.santhosh@btech.christuniversity.in	8848777178	ALB
10	2462014	ALLEN GEO PRASAD	4BTAD - AI & DS	allen.prasad@btech.christuniversity.in	7907572709	ALL
11	2462015	AMAN GAUTAM	4BTAD - AI & DS	aman.gautam@btech.christuniversity.in	9006935293	AMA
12	2462017	ANDREW SAJI KOVOOR	4BTAD - AI & DS	andrew.kovoor@btech.christuniversity.in	9645934571	AND
13	2462018	ANGEL MARIYA SHIBU	4BTAD - AI & DS	angel.shibu@btech.christuniversity.in	9746358178	ANG
14	2462019	APOORV RATHI	4BTAD - AI & DS	apoorv.rathi@btech.christuniversity.in	8445100000	APO
15	2462021	AVELYN ROXANA J	4BTAD - AI & DS	avelyn.j@btech.christuniversity.in	9445887224	AVE
16	2462022	AYMAN NAZIR CHOWDRY	4BTAD - AI & DS	ayman.chowdry@btech.christuniversity.in	7006850403	AYM
17	2462025	B.V.K.PREETHAM REDDY	4BTAD - AI & DS	b.reddy@btech.christuniversity.in	7842603848	PRE
18	2462026	BASIPOGU SAHAJ BHUVAN	4BTAD - AI & DS	basipogu.bhuvan@btech.christuniversity.in	8309100096	BAS
19	2462027	BATCHU YASWANTH SAI VARDHAN	4BTAD - AI & DS	batchu.vardhan@btech.christuniversity.in	8919694432	BAT
20	2462030	BUDDHA VIRESH REDDY	4BTAD - AI & DS	buddha.reddy@btech.christuniversity.in	9014167664	BUD
21	2462032	CHARAN H P	4BTAD - AI & DS	charan.p@btech.christuniversity.in	9611596700	CHA
22	2462035	D R DARSHAN KUMAR	4BTAD - AI & DS	d.kumar@btech.christuniversity.in	8971714571	DAR
23	2462036	DEVANSH CHHAWCHHARIA	4BTAD - AI & DS	devansh.chhawchharia@btech.christuniversity.in	9330960012	DEV
24	2462040	G JONATHAN DANIEL	4BTAD - AI & DS	g.daniel@btech.christuniversity.in	7305988358	JON
25	2462042	GAJJALA SNEHITHA REDDY	4BTAD - AI & DS	gajjala.reddy@btech.christuniversity.in	7702812288	GAJ
26	2462043	HARSHDEEP SINGH	4BTAD - AI & DS	harshdeep.singh@btech.christuniversity.in	9451107572	HAR
27	2462044	HIMANSHU AGRAWAL	4BTAD - AI & DS	himanshu.agrawal@btech.christuniversity.in	9520444390	HIM
28	2462046	HITESH SAI SHARMA G V	4BTAD - AI & DS	hitesh.v@btech.christuniversity.in	9606822238	HIT
29	2462049	ISHAN GAYEN	4BTAD - AI & DS	ishan.gayen@btech.christuniversity.in	8100146059	ISH
30	2462051	J H K CHARAN VARMA	4BTAD - AI & DS	j.varma@btech.christuniversity.in	7995837130	CHA
31	2462052	J S REVANTH KUMAR REDDY	4BTAD - AI & DS	j.reddy@btech.christuniversity.in	7842777179	REV
32	2462055	JESSICA S RAJ	4BTAD - AI & DS	jessica.raj@btech.christuniversity.in	7676646561	JES
33	2462086	JINKALA SHAHANAZ	4BTAD - AI & DS	jinkala.shahanaz@btech.christuniversity.in	8639527123	JIN
34	2462090	JYOTHIR MAI GUMMADI	4BTAD - AI & DS	jyothir.gummadi@btech.christuniversity.in	9030206927	JYO
35	2462092	K S V AMITH VARMA	4BTAD - AI & DS	k.varma@btech.christuniversity.in	7330768333	AMI

36	2462094	K PRANAV KRISHNA	4BTAD - AI & DS	k.krishna@btech.christuniversity.in	8277259163	PRA
37	2462096	K.S.M.LAKSHMI NARAYANA REDDY	4BTAD - AI & DS	k.reddy@btech.christuniversity.in	9381862562	LAK
38	2462097	KALLURI S V RAJESWARI DEVI	4BTAD - AI & DS	kalluri.devi@btech.christuniversity.in	7981881699	KAL
39	2462098	KANDASAMY G	4BTAD - AI & DS	kandasamy.g@btech.christuniversity.in	9442868933	KAN
40	2462099	KANISHK T N	4BTAD - AI & DS	kanishk.n@btech.christuniversity.in	6364024220	KAN
41	2462100	KANISHKA MEENA	4BTAD - AI & DS	kanishka.meena@btech.christuniversity.in	9413996843	KAN
42	2462101	KARASALA CHANDU	4BTAD - AI & DS	karasala.chandu@btech.christuniversity.in	9908127393	KAR
43	2462102	KARTHIK S REDDY	4BTAD - AI & DS	karthik.reddy@btech.christuniversity.in	9019623259	KAR
44	2462103	KEERTHANA S	4BTAD - AI & DS	keerthana.s@btech.christuniversity.in	8618641151	KEE
45	2462104	KETHAN M S	4BTAD - AI & DS	kethan.s@btech.christuniversity.in	7483664057	KET
46	2462105	KETHINI MANOHAR	4BTAD - AI & DS	kethini.manohar@btech.christuniversity.in	9000889100	KET
47	2462107	KONDURU GURU SAI TARUN	4BTAD - AI & DS	konduru.tarun@btech.christuniversity.in	8309177114	KON
48	2462108	KOTIPALLI MOHANA SURYA HARSHA	4BTAD - AI & DS	kotipalli.harsha@btech.christuniversity.in	8977017268	KOT
49	2462109	KOTRA REVANTH REDDY	4BTAD - AI & DS	kotra.reddy@btech.christuniversity.in	7893902999	KOT
50	2462110	M ELIJO PAUL	4BTAD - AI & DS	m.paul@btech.christuniversity.in	7418658638	ELI
51	2462111	M MOHITH REDDY	4BTAD - AI & DS	m.reddy@btech.christuniversity.in	8374246960	MOH
52	2462112	M V DEEKSHITHA REDDY	4BTAD - AI & DS	m.reddy2@btech.christuniversity.in	9392817299	DEE
53	2462113	M V S ANANT SAJIV	4BTAD - AI & DS	m.sajiv@btech.christuniversity.in	6383637175	ANA
54	2462115	M. CHARAN TEJA	4BTAD - AI & DS	m.teja@btech.christuniversity.in	9959648939	CHA
55	2462150	R JERPHIN	4BTAD - AI & DS	r.jerphin@btech.christuniversity.in	9442868933	JER
56	2462155	SARAN S	4BTAD - AI & DS	saran.s@btech.christuniversity.in	9876543210	SAR
57	2462160	VISHNU PRIYA M	4BTAD - AI & DS	vishnu.priya@btech.christuniversity.in	9876543211	VIS

3. Emergency SOS Panic Call System & Prank Call Rules

Emergency SOS Dispatch Workflow:

- Instant Activation:** Clicking 'Quick SOS Emergency' auto-pinpoints live GPS coordinates (lat, lng).
- Auto Student Profile Retrieval:** Fetches complete 57-field student data record from database.
- Simulated 2-Way Audio Line:** Opens live mic connection to Security Officer NNP.
- No Image & No 4 Steps for SOS:** SOS panic calls do **NOT** require live camera images and do **NOT** follow 4-step progress bars. They log directly to the **Dedicated SOS Log** table.
- Direct Call Confirmation:** Opens call confirmation modal with top-right **X** cross exit button and direct dial button: **CONFIRM & CALL SECURITY CONTROL (+91 8639527123)**.

Rules for Prank Callers & 4-Mark Disciplinary Lockout:

- Audio Warning:** Plays warning informing student that call is logged with device ID.
- Security Verification:** Security Guard NNP verifies each SOS call as either **[✓ Mark Genuine]** or **[■■■ Mark Prank SOS (+1 Mark)]**.
- Duplicate Penalty Protection:** Penalty marks cannot be issued twice for the same SOS call (penaltyApplied: true).
- Strict 4-Mark Threshold:** Reaching 4 Prank Marks sets accountStatus = 'DETAINED / ACCESS BLOCKED' and detentionStatus = 'Detained for 1 Year'.
- Full-Screen Website Blocking:** Detained students are locked out with a non-dismissible full-screen red warning overlay, active sessions are invalidated, and password resets are disabled.

4. Sequential 4-Step Incident Case Advancement System

Standard campus incident reports (Medical, Fire, Harassment, Cybercrime, Security Threat, Infrastructure) require WebRTC live camera photo/video proof and follow a strict sequential 4-step progress workflow managed by Security Officer NNP directly on card footers:

Step Index	Card Action Button Text	Status Code	System & Audit Action
Step 0	Update Step 1 ➔	REPORTED	Incident filed by student with mandatory WebRTC camera proof.
Step 1	Update Step 2 ➔	SECURITY APPROVED	Officer called Health Center / Expert Desk; step1 timestamp recorded.
Step 2	Update Step 3 ➔	SECURITY APPROVED	Paramedic / Security patrol team dispatched; step2 timestamp recorded.
Step 3	Update Step 4 ➔	SECURITY APPROVED	Response team arrived on spot and stabilized; step3 timestamp recorded.
Step 4	✓ Case Fully Resolved	RESOLVED	Final step complete; step4 timestamp recorded, case resolved, duration logged.

5. Campus Hotlines & Emergency Phone Registry

Emergency Category	Category Type	Priority Level	Dedicated Helpline Number
■ Security Control Desk	Primary Campus Hotline	HIGH	+91 8639527123
■ Medical Emergency / Ambulance	Health & Medical Desk	HIGH	+91 9442868933
■ Fire Emergency / Rescue	Fire Safety & Rescue	HIGH	+91 9442868933
■■ Harassment / Ragging Cell	Anti-Ragging Squad	HIGH	+91 9442868933
■ Cybercrime / Online Threat	Cyber Security Cell	MEDIUM	+91 9442868933
■ Accident / Severe Injury	Trauma & Ambulance	HIGH	+91 9442868933
■■ Suspicious Activity	Security Patrol	MEDIUM	+91 8639527123
■ Infrastructure Hazard	Facilities & Maintenance	LOW	+91 8639527123

6. Administration Portal & Active Universal Search Engine

Administration Analytics & Search Capabilities:

- **Live Global Search Input Bar (searchQuery):** Located at the top right of the tab navigation bar. Filters data in real-time across all 5 administrative tabs:
 - **Registry Tab:** Filters students by name, reg number, email, phone, class, and department.
 - **SOS Log Tab:** Filters SOS call logs by SOS ID, student name, phone number, verification status, and handling officer.
 - **Penalties Tab:** Filters students by prank mark count, active warnings, and detention status.
 - **Audit Trail Tab:** Filters incident audit trail by incident ID, category, student details, location, and status.
 - **Analytics Tab:** Renders a live search summary banner counting all matching entries across the system.
- **Complete Incident Audit Trail:** Logs exact ISO timestamps for step1, step2, step3, and step4 advancement and calculates total resolution duration in minutes.

7. Mobile QR Code System (Zero-Installation PWA)

The system includes an integrated Mobile QR Code modal accessible from student, security, admin, and login screens. Bound to local network host **http://172.16.208.138:3000** or **http://localhost:3000**, scanning the QR code opens the full PWA campus safety app on any mobile phone camera without downloading apps from Play Store or App Store.